

ORIGINAL ARTICLE

Gestão e Controlo de Qualidade de Caixas
Management and Quality Control of Boxes
Gestión y Control de Calidad de Cajas

RESUMO

Introdução: A Gestão da Qualidade das Caixas utilizadas em processos logísticos tem um grande impacto para a integridade dos produtos durante o transporte e armazenamento. A identificação de caixas danificadas e a monitorização do estado em geral das caixas pode prevenir perdas e otimizar a utilização de recursos. Assim, tendo como base esta relevância, este projeto sugere uma solução para essa gestão e monitorização, utilizando várias tecnologias para detetar o estado da caixa, ou seja, se esta se encontra danificada ou em bom estado.

Objetivo: O principal foco é automatizar o processo de controlo das condições da caixa, assegurando que a identificação de defeitos ou danos é mais rápida, assim como a sua separação.

Métodos: Os métodos abordados para o desenvolvimento desse projeto é com base em tecnologias que treinam um modelo de caixas, e com base nesse modelo identifica em que estado se encontra a caixa capturada por uma fotografia. Com base neste processo, a plataforma fornece dados detalhados sobre a caixa em questão, e o horário de monitorização da caixa, que posteriormente são enviados para uma base de dados, e tratados numa página *Web*.

Resultados: O projeto permitiu a implementação e desenvolvimento de um programa que identifica com uma taxa de precisão de acordo com o modelo de caixas, o estado das caixas em papelão, indicando se é uma caixa danificada, em bom estado ou se não detetou nenhuma caixa. Além disso, o *dashboard* concedeu de uma forma mais apelativa e organizada, uma visão geral do número de caixas relacionadas a cada estado.

Conclusão: Este projeto demonstrou ser eficaz para a gestão e monitorização das caixas, proporcionando à empresa uma forma precisa e automatizada de identificação de caixas danificadas. A implementação deste sistema ofereceu uma capacidade de monitorização em tempo real e a visualização de dados concretos. Assim, melhorou o controlo logístico e contribuiu para a redução de custos e otimização dos recursos, sendo possível a expansão para outras áreas de operação.

Palavras-chave: Caixas, Monitorização, Controlo, Gestão, Estado

ABSTRACT

Introduction: The Quality Management of Boxes in logistics significantly impacts product integrity during transportation and storage. Identifying damaged boxes and monitoring their condition can prevent losses and optimize resources. This project proposes a solution for managing and monitoring boxes using technologies to detect their condition, identifying whether they are damaged or in good condition.

Objective: Automate the process of monitoring box conditions, ensuring faster identification of defects or damages and facilitating their separation.

Methods: The project utilizes technologies to train a model based on box conditions. Using this model, the condition of a box captured in a photograph is identified. The platform provides detailed data on the box and its monitoring time, which are sent to a database and processed on a web page.

Results: A program was developed to identify the condition of cardboard boxes with high accuracy, classifying them as damaged, in good condition, or not detected. A dashboard presents a clear and organized overview of the number of boxes in each condition.

Conclusion: This project proved to be effective for the management and monitoring of boxes, providing the company with a precise and automated method for identifying damaged boxes. The implementation of this system offered real-time monitoring capabilities and the visualization of concrete data. Thus, it improved logistical control and contributed to cost reduction and resource optimization, with the potential for expansion to other operational areas.

Keywords: Boxes, Monitoring, Control, Management, Condition

RESUMEN

Introducción: La Gestión de la Calidad de las Cajas utilizadas en procesos logísticos tiene un gran impacto en la integridad de los productos durante el transporte y almacenamiento. La identificación de cajas dañadas y el monitoreo del estado general de las cajas puede prevenir pérdidas y optimizar el uso de recursos. Así, basándose en esta relevancia, este proyecto sugiere una solución para esta gestión y monitoreo, utilizando varias tecnologías para detectar el estado de la caja, es decir, si se encuentra dañada o en buen estado.

Objetivo: El principal enfoque es automatizar el proceso de control de las condiciones de la caja, asegurando que la identificación de defectos o daños sea más rápida, así como su separación.

Métodos: Los métodos abordados para el desarrollo de este proyecto se basan en tecnologías que entrenan un modelo de cajas, y con base en este modelo, identifican en qué estado se encuentra la caja capturada en una fotografía. A partir de este proceso, la plataforma proporciona datos detallados sobre la caja en cuestión y el horario de monitoreo de la caja, que posteriormente son enviados a una base de datos y procesados en una página web.

Resultados: El proyecto permitió la implementación y desarrollo de un programa que identifica, con una tasa de precisión según el modelo de cajas, el estado de las cajas de cartón, indicando si es una caja dañada, en buen estado o si no se detectó ninguna caja. Además, el panel proporcionó de manera más atractiva y organizada una visión general del número de cajas relacionadas con cada estado.

Conclusión: Este proyecto demostró ser eficaz para la gestión y monitoreo de las cajas, proporcionando a la empresa una forma precisa y automatizada de identificación de cajas dañadas. La implementación de este sistema ofreció una capacidad de monitoreo en tiempo real y la visualización de datos concretos. Así, mejoró el control logístico y contribuyó a la reducción de costos y optimización de recursos, siendo posible la expansión a otras áreas de operación.

Palabras Clave: Cajas, Monitoreo, Control, Gestión, Estado

INTRODUCTION

In the business context, the management and quality control of boxes plays a fundamental role in product safety and customer satisfaction. Boxes are an important element in the supply chain and in global trade, whether for storage, transport or presentation. However, there are several factors that can generate additional costs, waste of resources and damage to the company's reputation, such as lack of standardization, the use of inappropriate materials or failure to verify integrity.

This article addresses the main challenges and methodologies used in the management and quality control of boxes, with a focus on verification model training methods and emerging technologies.

1. THEORETICAL FRAMEWORK, LITERATURE REVIEW, STATE OF THE ART, CONCEPTUAL MODEL

Product quality management and control promises operational efficiency, process control and product quality assurance.

The quality management of a product (in this case, the boxes) is divided into three fundamental points, planning, improvement and quality control, defended by Joseph M. Juran, in his book *Managerial Breakthrough*, where he places quality improvement at the top of a manager's priorities. The quality of a product must be integral in all production processes and not just in the final stages of inspection.

Several studies have explored the quality control of products, focusing mainly on methods such as visual inspections, resistance tests, emerging technologies and statistical control tools in order to have a more detailed view of the entire production line, conducting studies on the results obtained.

In recent years, industries have adopted several innovations in order to improve product management and quality control, implementing more precise technologies such as the use of smart sensors, cameras and data analysis.

In this project, a study was carried out with the objective of analyzing boxes, identifying damaged boxes, boxes in good condition and absence of boxes along a production line. The model is focused on three main components:

- **Continuous Monitoring:** Use of a camera in order to capture images of boxes in real time, in order to accurately analyze their characteristics.
- **Statistical Process Control:** Training of the model to compare the images obtained, in order to identify possible inconsistencies.
- **Data analysis and management:** Analyze and manage the results obtained throughout the image captures, carrying out several studies on them.

2. METHODS

2.1 Detailed Process Description

The development of the project was based on the use of a *Raspberry Pi* with a camera to capture the images, due to its simplicity, low cost and versatility in integration with different technologies and sensors. To train the detection model, images of boxes were collected from *datasets* available *online*, submitted to a pre-processing process. The model was trained and evaluated with appropriate metrics, presenting satisfactory accuracy. The solution was integrated through an application that captures images of the boxes, processes them to identify the condition of the box (damaged or in good condition) and sends the data collected, such as the status and monitoring time. Subsequently, this information was processed and organized in a *dashboard* hosted on a *web* page, allowing for appealing and real-time monitoring.

2.1.1 Câmara do *Raspberry*

On the *Raspberry Pi*, the camera plays the role of capturing images in real time, which are then processed by the system. To enable this feature, a *Flask* server was implemented that acts as an HTTP access point (*endpoint*) for capturing images.

The developed code uses the *Picamera2* library to control the camera and configure the necessary settings, such as the resolution of the images (640x480). The *endpoint/capture* is responsible for capturing an image at the moment it is accessed, converting it to JPEG format using the *OpenCV* library, and sending the image as a response to any device on the network that requests it. This *endpoint* allows other components of the system, such as the computer containing the trained model, to obtain the captured images in a simple and efficient way.

This system turns the *Raspberry Pi* into a *capture interface* that communicates directly with the *project's processing and monitoring pipeline*.

2.1.2 Dataset, Google Colab e YOLOv8

For the model training, we used a *dataset* from *Roboflow*, a popular platform for computer vision projects. This *dataset* was chosen because it contains images of boxes in different states (damaged and in good condition), making it easier to identify and categorize during model training.

Google Colab was the tool chosen to train the model due to several factors, such as free access to *GPU* resources, which significantly speed up training, the ease of installation and integration of libraries, and support for code in *Python*, which was the language used in the project. The combination of an accessible and powerful environment was crucial to complete this step without the need for *robust local hardware*.

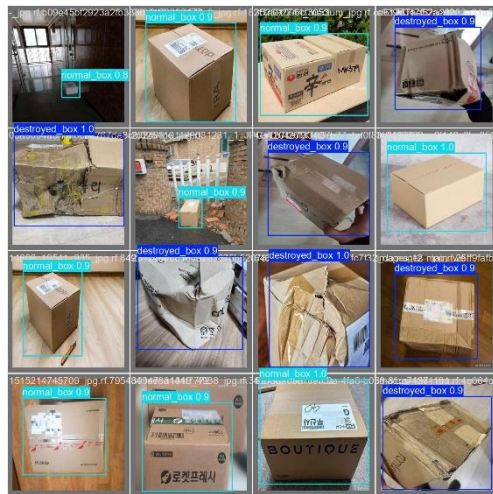


Figure 1 - Model

A picture with text, card, carton, Merchandise box Automatic description YOLOv8 was chosen as the machine learning model due to its reputation for object detection tasks. It offers a great combination of accuracy and efficiency, making it ideal for projects that need fast and effective processing. YOLOv8 allowed it to identify and classify boxes as "normal box" and "destroyed box" with high accuracy.

In Figure 1, it is possible to see the results of the YOLOv8 model. Each box in the image was identified and categorized as "normal box" or "destroyed box". These labels represent the state of the box, with an associated score, which is the confidence of the model in the classification performed.

2.2 Technologies and Tools Used

In this project, the following tools were used, the *Raspberry Pi 3* (Figure 2) together with the camera, an application developed in *Python* and a *Web page* developed in *OutSystems*. So, for the *Raspberry Pi 3*, the operating system used was the *legacy 32-bit Raspberry Pi OS* and for the camera to work properly, we used the *Therenny IDE*. For the application developed in *Python*, the *IDE*, *Visual Studio Code*, was used. The model used to identify the boxes was trained through *Google Colab* and *YOLO* version 8 and the *dataset* was taken from the *Roboflow website*.



2.3 Graphical Analysis

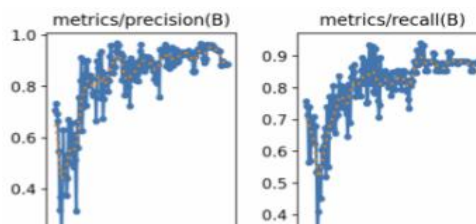


Figure 3 - Model Charts

Regarding the graphical analysis of the training of the *YOLO* model for the classification of boxes in good condition and damaged boxes, in Figure 3 two graphs are presented that indicate the evolution of the metrics of accuracy (left) and recall (right). A trend of improvement in both metrics is observed throughout the training, indicating that the model is learning to distinguish the different classes. Precision 0.9 shows the ability of the model to make consistent predictions. The initial oscillations are attributed to the adjustment of weights and variations in the validation data.

3. RESULTS

The results obtained from this project were recorded and analyzed on the *Outsystems platform* (Figure 4), where it is possible to view the entire history of the captured images, status, validation date and other information on a *Web page* developed by the platform. On this same page it is possible to analyze various information that will be covered in the following topics:

3.1 Image History and Validation

On the platform it is possible to view a list with all the images captured during the production process, with a more detailed view including their status.

For each box, the date and time that the validation was performed is recorded, which helps to identify fault behaviors over time.

3.2 Sorting of Crates

The system was able to classify the condition of the boxes into the intended categories: Damaged Boxes, Boxes in Good Condition and Uncertain Outcome

For each box status, its total and percentage are presented, thus having a broader and more intuitive view of the quality of the boxes.

3.3 Real-Time Monitoring

By collecting data and using a *real-time dashboard* (Figure 4), it allows production managers to continuously monitor the quality of the boxes, thus quickly adjusting any inconvenience.

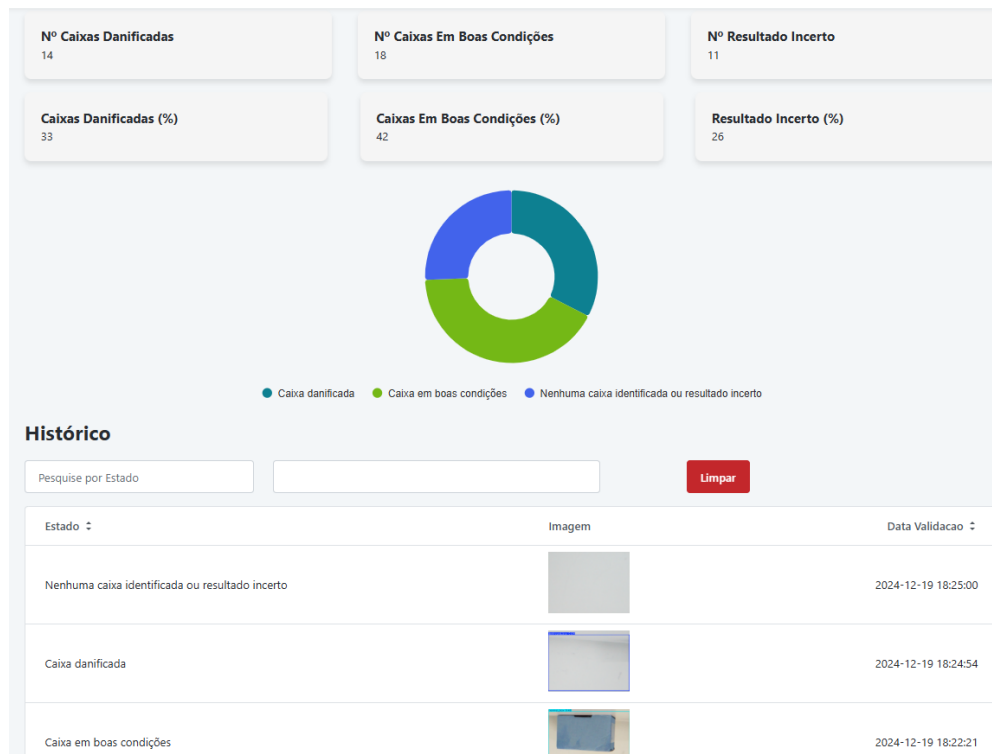


Figure 4 - Outsystems Results

4. DISCUSSION

4.1. Implication of Results Obtained

The results obtained are influenced by the quality of the components used. Thus, in the case of capturing the images, the problems encountered were the poor image quality of the camera used and the little processing offered by the *Raspberry Pi 3*. Another problem was the quality of the model trained to identify the boxes and their damage, in which sometimes the identifications were incorrect or of little certainty. On the other hand, the *dashboard* developed shows the data clearly and represents one of the strengths of the project.

4.2. Future Improvements

As future improvements, there is the possibility of integrating new viewing angles on the boxes and improving the model that detects the boxes and their damage. We can also improve the equipment used, that is, use a computer with greater processing capacity, as well as a camera with better resolution quality. Finally, other parameters for measuring and detecting defects in the boxes can be used, such as weight.

CONCLUSION

In short, the project revealed the effectiveness of an automated system for identifying and monitoring the quality of boxes, helping to improve a company's logistics. The realization of the system together with

the *dashboard* allowed the company to better analyze statistical data in real time, ensuring greater efficiency in the storage process.

Thus, the results will indicate to the company whether there was a reduction in damaged boxes compared to previous months, which impacts on better management and optimization of resources.

The initially proposed objectives were partially fulfilled, only the implementation of the logic for the possibility of validating the weight of the box was not carried out, due to the incompatibility with the material used (*Raspberry*). Throughout the development of the project some doubts and difficulties also arose, however they were overcome through several researches and study of the functionalities.

Overall, this project was quite intuitive and important for all members of the group, as a lot of knowledge and practices were acquired with these new technologies.

This project has the possibility of being expanded and adapted to other areas of the company, enabling a broader and more robust solution, and a more solid base. An example of expansion would be to cover other types of packaging or logistics equipment, such as transport containers, among others.

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CONFLICT OF INTEREST

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